Dhruv Malhotra

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SUMMARY

Software Engineer in Test with 6+ years building enterprise-scale test automation, API validation frameworks, and performance benchmarking tools across cloud-native platforms (AWS, Docker, Kubernetes). Proven expertise in Python backend engineering, high-coverage testing, and CI/CD pipelines for SaaS products serving millions of users. Security clearance eligible veteran with a track record of bridging QA and backend engineering to drive system reliability and business outcomes.

SKILLS

Programming/ML: Python, Java, Django, PyTorch, Pandas, TensorFlow, Flask, React.js, C/C++, Swift
Cloud/Tools: AWS (EC2, Lambda, S3, CloudWatch), Docker, Kubernetes, Jenkins, GitHub Actions, Git, Postman, MySQL, PostgreSQL, Redis, Browserstack
QA/Testing: pytest, pytest-django, Selenium, Playwright, Postman, JUnit, k6, BDD, TDD, JIRA, REST API Testing, Performance Testing
Monitoring: Grafana, DataDog, Prometheus

WORK EXPERIENCE

Software Engineer in Test, Brivo (Formerly Eagle Eye Networks) Austin TX | October 2024 – Present

• Built a multi-cluster synthetic monitoring platform (Python/Flask) that continuously validates event-driven notification pipeline integrity across 24 production environments in 6 global regions via automated event injection, reconciliation, SLA compliance tracking, and email delivery verification — serving 39+ API routes with real-time health dashboards
• Developed an LLM-augmented QA workflow that automated API test generation and data analysis, reducing manual testing cycles from days to ~15 minutes across enterprise API endpoints
• Designed and executed end-to-end test suites for a distributed error-handling and retry system, validating cooloff behavior, retry policies, and webhook concurrency across RESTful APIs
• Analyzed 3,100+ alerts and 10,100+ notifications to reverse-engineer undocumented throttling behavior, discovering a shared rule-level cooloff mechanism that directly influenced platform architecture decisions
• Built a fault-tolerant Gmail ingestion pipeline to batch-process 200+ emails/day, enrich alert data via API, and export to CSV; improved throughput 5× using concurrency, caching, and backoff logic
• Developed a high-concurrency API load testing tool (Python/aiohttp) with session-based auth, dynamic CLI filtering, pagination handling, and Docker CI/CD integration for benchmarking core platform APIs
• enchmarked core platform APIs (alerting, notifications, rules management), automating P95/P99 latency, throughput, and error-rate metrics to establish performance baselines and surface filter-driven bottlenecks
• Authored 1,000+ unit tests (pytest-django) and 300+ Postman API tests, achieving 100% coverage across core services and validating authentication, role-based permissions, and complex end-to-end workflows
• Conducted an API input-validation audit across 6+ endpoint groups, uncovering a critical PATCH vulnerability in a rules endpoint that allowed removal of required fields in production — led to immediate remediation
• Validated a platform migration impacting 10,000+ users; developed automated comparison scripts that identified and resolved a 15% performance discrepancy, confirming 99.9%+ delivery reliability post-migration

Software Engineer, Yunex Traffic Austin TX | July 2023 – October 2024

• Developed real-time traffic management software in Python, integrating complex scheduling algorithms and database-backed state handling, improving urban traffic flow efficiency 15% across multiple city networks
• Integrated AI/ML models with TensorFlow and Python for predictive traffic analysis, applying time-series forecasting and anomaly detection to cut peak-hour congestion 20% and reduce operational costs 30%
• Migrated legacy traffic management systems to AWS, redesigning services for cloud-native deployment and leveraging Docker + Kubernetes clusters to modernize infrastructure, improve scalability, and system reliability

• Built a load testing framework simulating 3,000+ intersections, incorporating concurrency modeling and data-driven traffic scenarios to validate reliability and scalability of large-scale traffic management systems
• Designed automated test suite validating real-time traffic signal coordination across 50+ intersections, achieving 95%+ timing accuracy under variable load conditions
• Collaborated with embedded systems team to validate NTCIP protocol compliance across 200+ traffic controllers, reducing field deployment failures by 30%

Software Engineer, Givelify Austin TX | May 2020 – July 2023
• Developed and deployed a recommendation engine using Python and PyTorch, elevating user retention by 25% and engagement by 20% through cross-functional collaboration with engineering, product, and quality teams
• Refined search capabilities using Python and React, benefiting 10k+ daily users by boosting search efficiency by 15%
• Implemented geocoding services using Google API and LocationIQ, improving location-based search accuracy for 5,000+ daily users and reducing failed lookups by 40%
• Improved donation workflows by building new API endpoints and streamlining payment integrations, reducing transaction latency by 18% and enhancing reliability of high-volume transactions for thousands of daily users
• Built CI/CD pipelines with Jenkins and Docker, cutting deployment time by 70% and improving release reliability
• Architected and maintained pytest test suite with 85%+ code coverage across donation processing, payment gateway, and user authentication services
• Led migration of monolithic payment service to microservices architecture, improving transaction throughput by 45% and reducing latency by 25%

Junior Software QA Developer, Nourtek Dallas, TX | October 2019 – May 2020
• Developed and executed 500+ automated JUnit test cases across 3 major releases, reducing production defects by 20%
• Piloted regression testing using Jira, cementing software stability across 25+ releases
• Built automated regression test framework in Java/JUnit, reducing manual QA cycle from 3 days to 4 hours

Navy Corpsman, United States Navy Okinawa, JP | July 2011 – July 2016
• Provided medical care for 4,500+ servicemembers in high-stakes operational environments; supported S-2 Intelligence Division managing security clearances for 3,000+ personnel
• Developed leadership, mission-critical decision-making, and cross-functional coordination skills in a fast-paced military environment

EDUCATION

Bachelor of Science in Computer Science University of Texas at Dallas, Richardson, TX | Aug. 2018 - Dec 2021
Associate of Science in Computer Science Austin Community College, Austin, TX | Aug. 2016 - Aug 2018